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Retrieval-Augmented Generation

In this example, we will build and test a simple Retrieval-Augmented Generation with Large Language Models (RAG-LLM). It typically has two main components:

- Retriever identifies relevant context or documents from an external data source. It often uses embeddings to find and rank content based on similarity to the user's query. Popular retrievers include dense retrievers such as Sentence Transformers, FAISS, and BM25-based systems. The retriever enables RAG-LLM to pull in precise, contextual information to aid the generative model in crafting accurate responses.
- Generators usually use a large language model, such as GPT or T5, which takes the retrieved context and the user's query as input and generates a coherent, contextually relevant answer. The generator relies on the retriever-provided context to produce detailed, accurate answers to specific questions.

The example below includes,

- The external data source is from the JSON file `squad-train-v2.0.json`, which is part of the SQuAD (Stanford Question Answering Dataset) v2.0, a popular dataset for training and evaluating machine learning models on reading comprehension and question-answering tasks. Stanford released this dataset and is widely used in natural language processing (NLP) for benchmarking question-answering systems (source: 📖 **https://rajpurkar.github.io/SQuAD-explorer/**)
- Retriever is a `Sentence Transformer` (source: 🤗 **https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2**)
- Large Language Model (LLM) is a `Llama-3.2-1B-Instruct` model (source: ♾ **https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct**)

▶️ Install the sentence transformer: `pip install sentence-transformers` ↩

Meta requires permission to download the model, which requires an `edu` email address. As in the previous lectures, follow the instructions to request permission.

Since the LLM we utilize in this example, `meta-llama/Llama-3.2-1B-Instruct`, can process the entire context, we will store only two key pieces of information: topics and contexts. At this time, we will not process the `qas` section (question-answer section).

In [1]:
```
%matplotlib inline
import matplotlib.pyplot as plt
plt.rcParams['figure.dpi'] = 72
from IPython.display import Markdown
import numpy as np
import pandas as pd
import os
import json
from sklearn.metrics.pairwise import cosine_similarity


MODEL_PATH= '/EP_models/'
os.environ['HF_HOME'] = MODEL_PATH  # before import transformers
os.environ['HF_DATASETS_OFFLINE']= '1'
```

In [2]:
```
import torch
import transformers
from transformers import pipeline
from sentence_transformers import SentenceTransformer
from transformers import AutoModelForCausalLM, AutoTokenizer

# filter warnings
import warnings
transformers.logging.set_verbosity_error()
warnings.simplefilter(action='ignore', category=FutureWarning)
warnings.simplefilter(action='ignore', category=UserWarning)

Device = torch.device("cuda:0" if torch.cuda.is_available() else "cpu")

print(f'PyTorch version= {torch.__version__}')
print(f'transformers version= {transformers.__version__}')
print(f'CUDA available= {torch.cuda.is_available()}')
```

```
PyTorch version= 2.9.1+cu130
transformers version= 4.57.6
CUDA available= True
```

In [3]:
```
# from huggingface_hub import login
# login()
```

In [4]:
```
# explore SQuAD training data structure
with open('/EP_datasets/squad-train-v2.0.json', 'r') as f:
    squad_data = json.load(f)

n_titles = len(squad_data['data'])
print(f'There are {n_titles} titles/topics in SQuAD v2.0 training dataset.

title_idx = 200
```

```
prg_idx = 2
current_title = squad_data['data'][title_idx]['title']
n_paragraphs = len(squad_data['data'][title_idx]['paragraphs'])
n_qas = len(squad_data['data'][title_idx]['paragraphs'][prg_idx]['qas'])
print(f"Title with index #{title_idx} is {squad_data['data'][title_idx]['t
print(f"Title {current_title} has {n_paragraphs} paragraphs. Paragraph wit

# docs = [entry['context'] for entry in squad_data['data'][0]['paragraphs'
# squad_data['data']
```

```
There are 442 titles/topics in SQuAD v2.0 training dataset.

Title with index #200 is Florida.
Title Florida has 35 paragraphs. Paragraph with index 2 has 10 question-answ
er pairs.
```

In [5]:

```
# squad_data['data'][200]['paragraphs'][0]['qas']
squad_data['data'][200]['paragraphs'][0]['context']
```

Out[5]:

```
'Florida i/'flɒrɪdə/ (Spanish for "flowery land") is a state located in the
southeastern region of the United States. The state is bordered to the west
by the Gulf of Mexico, to the north by Alabama and Georgia, to the east by
the Atlantic Ocean, and to the south by the Straits of Florida and the sove
reign state of Cuba. Florida is the 22nd most extensive, the 3rd most popul
ous, and the 8th most densely populated of the United States. Jacksonville
is the most populous city in Florida, and the largest city by area in the c
ontiguous United States. The Miami metropolitan area is the eighth-largest
metropolitan area in the United States. Tallahassee is the state capital.'
```

In [6]:

```
sqdata = {"topic": [], "context": []}

for i in range(n_titles):
    n_paragraphs = len(squad_data['data'][i]['paragraphs'])
    for j in range(n_paragraphs):
        sqdata["topic"].append(squad_data['data'][i]['title'])
        sqdata["context"].append(squad_data['data'][i]['paragraphs'][j]['c


df = pd.DataFrame(sqdata)

# sanity
print(f'Shape of df: {df.shape}')
df.head()
```

```
Shape of df: (19035, 2)
```

Out[6]:

	topic	context
0	Beyoncé	Beyoncé Giselle Knowles-Carter (/biːˈjɒnseɪ/ b...
1	Beyoncé	Following the disbandment of Destiny's Child i...
2	Beyoncé	A self-described "modern-day feminist", Beyonc...
3	Beyoncé	Beyoncé Giselle Knowles was born in Houston, T...
4	Beyoncé	Beyoncé attended St. Mary's Elementary School ...

In [7]:
```
class RAGLLM:
    def __init__(self, external_data):
        self.st_model = SentenceTransformer('all-MiniLM-L6-v2')  # sentenc

        # external data
        self.external_data = external_data
        self.titles = self.external_data['topic'].unique()
        self.title_embeddings = self.st_model.encode(self.titles.tolist())

        # LLM model
        self.generator = pipeline('text-generation', model='meta-llama/Lla
                                  max_length=4096, device=Device)  # utili

    def generate_answer(self, user_question):
        # Step 1: Find the best matching topic
        question_embedding = self.st_model.encode([user_question])
        title_similarities = cosine_similarity(question_embedding, self.ti
        best_title_index = np.argmax(title_similarities)
        best_title = self.titles[best_title_index]
        print(f"The best topic: {best_title}\n")

        # Step 2: Filter contexts for the identified best topic
        topic_contexts = self.external_data[self.external_data['topic'] ==

        # Step 3: Vectorize each context under the best topic and find the
        context_embeddings = self.st_model.encode(topic_contexts)
        context_similarities = cosine_similarity(question_embedding, conte
        best_context_index = np.argmax(context_similarities)
        best_context = topic_contexts[best_context_index]
        print(f"The best paragraph: {best_context}\n")

        # Step 4: Generate the answer using LLM with retrieved context
        prompt = f"Context: {best_context}\n\nQuestion: {user_question}\nA
        response = self.generator(prompt, max_length=300)
        answer_text = response[0]['generated_text']
        answer = answer_text.split("Answer: ")[-1]
        return answer
```

In [8]:
```
query = "Who is the US president in 2012?"
ragllm = RAGLLM(df)
output = ragllm.generate_answer(query)
for part in output.split("\n\n"):
    print(part)
```

```
The best topic: United_States_presidential_election,_2004

The best paragraph: Just eight months into his presidency, the terrorist att
acks of September 11, 2001 suddenly transformed Bush into a wartime presiden
t. Bush's approval ratings surged to near 90%. Within a month, the forces of
a coalition led by the United States entered Afghanistan, which had been she
ltering Osama bin Laden, suspected mastermind of the September 11 attacks. B
y December, the Taliban had been removed as rulers of Kabul, although a long
and ongoing reconstruction would follow, severely hampered by ongoing turmoi
l and violence within the country.

Barack Obama
The correct answer is Barack Obama. He was inaugurated as the 44th President
of the United States in January 2009, not 2012.
```

Analysis: The output is correct. The best topic from the external data source appears to be relevant, although the year in the best topic is 2004, while the question asks about 2012. The best paragraph mentions only President Bush, but the answer is Barack Obama which is correct.

In [9]:

```python
query = "What is the largest state in USA by area?"
ragllm = RAGLLM(df)
output = ragllm.generate_answer(query)
for part in output.split("\n\n"):
    print(part)
```

```
The best topic: Geography_of_the_United_States

The best paragraph: By total area (water as well as land), the United States
is either slightly larger or smaller than the People's Republic of China, ma
king it the world's third or fourth largest country. China and the United St
ates are smaller than Russia and Canada in total area, but are larger than B
razil. By land area only (exclusive of waters), the United States is the wor
ld's third largest country, after Russia and China, with Canada in fourth. W
hether the US or China is the third largest country by total area depends on
two factors: (1) The validity of China's claim on Aksai Chin and Trans-Karak
oram Tract. Both these territories are also claimed by India, so are not cou
nted; and (2) How US calculates its own surface area. Since the initial publ
ishing of the World Factbook, the CIA has updated the total area of United S
tates a number of times.

Montana, with a total area of 147,040 square miles.
```

Analysis: The output is incorrect. The best topic from the external data source appears to be relevant. However, the best paragraph is quite off. It is also possible that the chosen topic wasn't the most relevant one. From the paragraph, we can see that the geography of the US basically discusses the geography of the US as a country, not its individual states. The information about the largest state in the US by area is possibly in the Alaska topic.

In [10]:
```
query = "Which state is larger by area in the USA: Alaska or Texas?"
ragllm = RAGLLM(df)
output = ragllm.generate_answer(query)
for part in output.split("\n\n"):
    print(part)
```

```
The best topic: Alaska

The best paragraph: Alaska (i/ə'læskə/) is a U.S. state situated in the nort
hwest extremity of the Americas. The Canadian administrative divisions of Br
itish Columbia and Yukon border the state to the east while Russia has a mar
itime border with the state to the west across the Bering Strait. To the nor
th are the Chukchi and Beaufort Seas, the southern parts of the Arctic Ocea
n. To the south and southwest is the Pacific Ocean. Alaska is the largest st
ate in the United States by area, the 3rd least populous and the least dense
ly populated of the 50 United States. Approximately half of Alaska's residen
ts (the total estimated at 738,432 by the Census Bureau in 2015) live within
the Anchorage metropolitan area. Alaska's economy is dominated by the fishin
g, natural gas, and oil industries, resources which it has in abundance. Mil
itary bases and tourism are also a significant part of the economy.

Alaska is larger by area. It covers approximately 663,268 square miles, whil
e Texas covers approximately 268,597 square miles. This makes Alaska more th
an three times the size of Texas.
```

Analysis: The output is correct. Both the best topic and the best paragraph from the external data source are on point.

In [11]:
```
query = "What is the population of Florida?"
ragllm = RAGLLM(df)
output = ragllm.generate_answer(query)
for part in output.split("\n\n"):
    print(part)
```

```
The best topic: Florida

The best paragraph: The United States Census Bureau estimates that the popul
ation of Florida was 20,271,272 on July 1, 2015, a 7.82% increase since the
2010 United States Census. The population of Florida in the 2010 census was
18,801,310. Florida was the seventh fastest-growing state in the U.S. in the
12-month period ending July 1, 2012. In 2010, the center of population of Fl
orida was located between Fort Meade and Frostproof. The center of populatio
n has moved less than 5 miles (8 km) to the east and approximately 1 mile
(1.6 km) to the north between 1980 and 2010 and has been located in Polk Cou
nty since the 1960 census. The population exceeded 19.7 million by December
2014, surpassing the population of the state of New York for the first time.

20,271,272
Year: 2015
Population growth rate: 7.82%
Population of Florida in 2010: 18,801,310
Population of Florida in 2012: 19,051,077
Population of Florida in 2014: 19,714,115
Population of Florida in 2015: 20,271,272
The population of Florida has been steadily increasing over the
```

Analysis: The output is correct. This question is quite general, as it does not mention a specific timeline. Therefore, the answer depends on how recent the external data source is. The last update to the SQuAD 2.0 dataset was in 2018. Hence, it makes sense that the output returned the population of Florida in 2015. Both the best topic and the best paragraph are relevant to the question.

In [12]:

```python
query = "Where is Mount Rushmore?"
ragllm = RAGLLM(df)
output = ragllm.generate_answer(query)
for part in output.split("\n\n"):
    print(part)
```

```
The best topic: Appalachian_Mountains

The best paragraph: In Pennsylvania, there are over sixty summits that rise
over 2,500 ft (800 m); the summits of Mount Davis and Blue Knob rise over 3,
000 ft (900 m). In Maryland, Eagle Rock and Dans Mountain are conspicuous po
ints reaching 3,162 ft (964 m) and 2,882 ft (878 m) respectively. On the sam
e side of the Great Valley, south of the Potomac, are the Pinnacle 3,007 fee
t (917 m) and Pidgeon Roost 3,400 ft (1,000 m). In West Virginia, more than
150 peaks rise above 4,000 ft (1,200 m), including Spruce Knob 4,863 ft (1,4
82 m), the highest point in the Allegheny Mountains. A number of other point
s in the state rise above 4,800 ft (1,500 m). Snowshoe Mountain at Thorny Fl
at 4,848 ft (1,478 m) and Bald Knob 4,842 ft (1,476 m) are among the more no
table peaks in West Virginia.

Mount Rushmore is located in the Black Hills of South Dakota, United States.
```

Analysis: The output is correct. However, the best topic and the best paragraph in the external dataset are not relevant.

Analysis and Observations

The main purpose of the example above is to demonstrate a simple implementation of the RAG with an LLM. While this approach is useful, there are areas for improvement. Key points include:

1. External Dataset Update: The SQuAD 2.0 training dataset was last updated in 2018, while the Llama-3.2-1B-Instruct model was last updated in September 2024. As a result, the pre-trained LLM contains more recent information and data.
2. Dataset Limitations: The external dataset SQuAD 2.0 training dataset may have limitations that can cause the LLM to rely on pre-trained data instead of the external dataset for some outputs. This reflects the constraints of relying on solely static external sources.
3. Efficiency of Updates: In practice, updating external datasets is more efficient and optimized compared to updating the entire pre-trained data in an LLM. This efficiency contributes to the growing popularity and practicality of RAG-LLMs.

Why RAG-LLM Is Better Than LLM Alone?

Using RAG-LLM is often better than using an LLM alone because RAG combines the strengths of both retrieval-based systems and generative language models. Here's why it can be advantageous:

1. Up-to-Date Knowledge: Accesses external, real-time data; LLMs rely on static, pre-training knowledge.
2. Improved Accuracy: Grounds responses in retrieved documents, reducing hallucinations.
3. Domain-Specific Flexibility: Tailored to specific domains by indexing relevant data.
4. Reduced Training Costs: Updates only the retrieval database, avoiding LLM retraining.
5. Transparency: Provides references or sources for answers, enhancing trust.
6. Cost Efficiency: Uses smaller models with retrieval for similar or better performance.

In the example below, we will illustrate Domain-Specific Flexibility. In the SQuAD dataset, under the 'Florida' topic, 'Florida' is mentioned with its meaning as 'flower' in Spanish. This information is quite specific and possibly unique. However, the pre-trained dataset

of LLaMA may be more general. Therefore, while an LLM may not be able to answer the question correctly, a RAG-LLM will be able to do so in the Explanation part.

In [13]:
```
class LLaMAQA:
    def __init__(self, model_name='meta-llama/Llama-3.2-1B-Instruct'):
        self.model_name = model_name
        print(f"Loading model '{model_name}'...")
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)
        self.model = AutoModelForCausalLM.from_pretrained(model_name)

    def ask_question(self, question, max_length=50, temperature=0.7, top_p
        """
        Ask a question and get a response from the model.
        Parameters:
            question (str): The question to ask.
            max_length (int): Maximum length of the response.
            temperature (float): Sampling temperature for randomness.
            top_p (float): Nucleus sampling parameter.
            top_k (int): Top-k sampling parameter.
        Returns:
            str: The model's response.
        """
        inputs = self.tokenizer(question, return_tensors="pt")

        # generate the response
        outputs = self.model.generate(
            inputs["input_ids"],
            max_length=max_length,
            temperature=temperature,
            top_p=top_p,
            top_k=top_k,
            do_sample=True
        )
        # decode and return the response
        return self.tokenizer.decode(outputs[0], skip_special_tokens=True)
```

In [14]:
```
llama_qa = LLaMAQA()

query = "What does 'Florida' mean in Spanish?"
response = llama_qa.ask_question(query)
print("Q:", query)
print("A:", response)
```

```
Loading model 'meta-llama/Llama-3.2-1B-Instruct'...
Q: What does 'Florida' mean in Spanish?
A: What does 'Florida' mean in Spanish? 'Florida' is derived from the Latin
word 'floridanus', which means 'flowery' or 'blooming'. The name likely orig
inated from the state's lush and beautiful landscapes, which
```

In [15]:
```
%%html
<style>
    table {margin-left: 0 !important;}
    p {font-family: verdana;}
    li {font-family: verdana;}
```

```
    div {font-size: 10pt;}
</style>
<!-- Display markdown tables left oriented in this notebook. -->
```